

Session: PIE CHART MASTERCLASS

"How to See the Whole Pizza"

Duration

60 Minutes

Step 1: The Pizza Story

Walk into class carrying (or drawing) a pizza.

Ask:

"If I cut a pizza into 8 slices and I eat 4 slices, how much pizza did I eat?"

Students:

50%

Perfect.

Now ask:

Why did you say 50%?

Because:

4 out of 8 slices.

Half.

Then write:

Pie Charts = Parts of a Whole

This is the most important idea.

Activity 1: Build a Human Pie Chart

Ask:

Favorite Subject

Count students.

Example:

Subject	Students
----------------	-----------------

Science	15
---------	----

Maths	10
-------	----

English	5
---------	---

Total = 30

Now calculate:

Science:

$$15/30 = 50\%$$

Maths:

$$10/30 = 33\%$$

English:

$$5/30 = 17\%$$

Ask:

Can we turn this into a pizza?

Yes.

Each subject gets a slice.

Why Pie Charts Exist

Imagine:

Subject	Students
Science	15
Maths	10
English	5

The table tells numbers.

But the pie chart instantly answers:

"Who owns the biggest share?"

Science.

Pie Chart Superpower

Write:

Bar Chart → Comparison

Pie Chart → Composition

Students must remember this.

Visual Example

Suppose class screen time:

Activity	Hours
YouTube	5
Instagram	3
Gaming	2

Total = 10 hours

Pie chart shows:

- YouTube = 50%
- Instagram = 30%
- Gaming = 20%

Students immediately see where time goes.

The Golden Rule

Ask:

When should we use a pie chart?

Students think.

Then teach:

Use pie charts when:

✓ Total matters

✓ Categories form one whole

✓ You want percentages

Examples:

- Budget spending
 - School house participation
 - Subject preferences
 - Mobile usage
-

When NOT To Use Pie Charts

This is where students become data literate.

Example

Fruit	Votes
Apple	50
Mango	49
Banana	48
Orange	47
Grapes	46
Guava	45
Pear	44

Pie chart becomes messy.

Too many slices.

Hard to compare.

Ask:

Which chart is better?

Bar chart.

Exactly.

Rule of Thumb

Write:

3–5 Categories

Pie chart is great.

More than 6 Categories

Bar chart is usually better.

Activity 2: School Budget Game

Suppose school spends:

Category	Amount
----------	--------

Teachers	50%
----------	-----

Building	20%
----------	-----

Sports	15%
--------	-----

Library	15%
---------	-----

Ask:

Where does most money go?

Teachers.

Pie chart answers instantly.

Understanding Percentages

Many students struggle here.

Use chocolate.

Imagine:

10 chocolates.

You eat:

5 chocolates.

How much?

50%

Eat:

2 chocolates.

How much?

20%

This makes pie charts easy.

Activity 3: Become a Pie Chart

Take 20 students.

Assign:

10 Science lovers

5 Maths lovers

5 English lovers

Make them stand in a circle.

Now physically create slices.

Students become the chart.

Pie Charts Can Lie Too

Very important.

Show:

Chart A

Proper percentages.

Chart B

Slices don't add to 100%.

Ask:

Is this possible?

No.

Pie Chart Detective Rule

Always ask:

Do all slices together equal 100%?

If not:

Something is wrong.

The 3D Pie Chart Trap

Ask:

Which slice looks bigger?

Front slice often looks larger.

Even if percentages are equal.

Tell students:

3D Pie Charts Look Cool

But They Can Distort Reality

Activity 4: Spot The Fake Pie Chart

Show:

- Missing labels
- Percentages > 100%
- Too many slices
- 3D distortion

Students become graph detectives.

Pie Chart vs Bar Chart

This is the most important comparison.

Bar Chart	Pie Chart
Compare values	Compare shares
Best for ranking	Best for percentages
Handles many categories	Best with few categories
More accurate	More intuitive

Classroom Challenge

Collect:

Favorite Sport

Convert to:

1. Table
2. Bar Chart
3. Pie Chart

Ask:

Which visualization answers which question?

Students discover:

Bar Chart:

Who has the most votes?

Pie Chart:

What percentage likes each sport?

AI Connection

Ask:

How do companies understand their customers?

Pie charts show:

- Age groups
- Device usage
- Purchase categories

AI teams use these charts to understand data before building models.

Final Thought Experiment

Ask:

If 70% of students spend more than 3 hours on social media...

What does that tell us about the whole class?

Students begin thinking in proportions.

This is exactly the mindset pie charts develop.

One Sentence Students Must Remember

Write dramatically:

"A Bar Chart compares categories."

"A Pie Chart shows how categories share a whole."

If they remember this distinction, they will already understand more about visualization than many adults.

Daily Screen Time Breakdown

How a student's 10 hours of screen time is divided.

activity	hours
YouTube	5
Instagram	3
Gaming	2